Feasibility Report

Analog Audio Effect Processor

Group Members:
230045X, 230140J, 230477X, 230730T

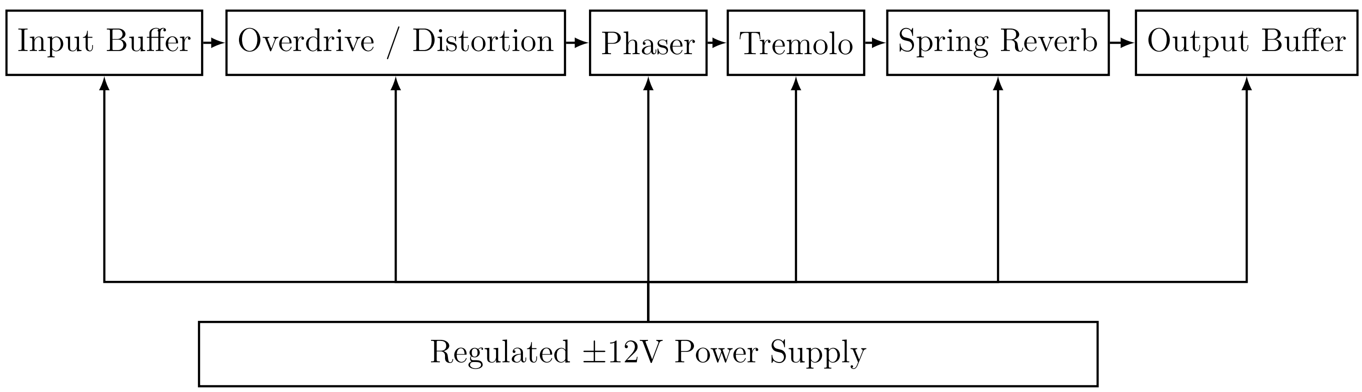
Functionality

The proposed **Analog Audio Effect Processor** is a modular, fully analog device designed to apply multiple sound effects to an audio signal in real time. The system will feature four distinct audio effects as follows:

- **Overdrive/Distortion** – Adds harmonic saturation and controlled clipping for a warm or aggressive tone.
- **Phaser** – Produces sweeping, phase-shifted comb filter effects using multiple all-pass filter stages.
- **Tremolo** – Modulates the amplitude of the signal periodically for a pulsing effect.
- **Spring Reverb** – Simulates reverberation using a mechanical spring tank.

Each module will be independently switchable, allowing the user to activate any combination of effects as the user wants. The signal will be processed in series for a straightforward and compact design.

Block Diagram of the System



Methodology

The development process is going to be carried out in the following stages:

1. **Research and Analysis:** Review existing analog effect circuits, studying their topologies, performance characteristics, and integration requirements. Identify component types suitable for low-noise, high-fidelity audio.
2. **Design Stage:**
   - Draw schematics for each effect module (Overdrive, Phaser, Tremolo, Spring Reverb) using professional software.
   - Incorporate buffers at the input and output of each effect for impedance matching.
   - Include individual bypass switching systems for each effect to allow any combination of active modules.
3. **Simulation:** Model each circuit in LTspice to verify frequency response, distortion levels, and modulation behavior before prototyping.
4. **Prototyping:**
   - Build each module separately on a breadboard.
   - Test each module's sound quality, noise level, and switching performance.
5. **Integration:** Interconnect modules in series on a unified breadboard prototype. Check that bypass switching works without pops or signal degradation.
6. **PCB Design:** Design the final PCB with careful grounding and power decoupling to minimize interference. Ensure physical layout accommodates control knobs and switches.
7. **Enclosure Design:** Create a SolidWorks 3D model for the enclosure, considering mechanical mounting for the spring reverb tank and ergonomic placement of controls.
8. **Assembly and Testing:** Assemble all modules on the PCB, install into the enclosure, and perform full system audio tests under different effect combinations.

Micro-products and Interconnections

The system consists of four main effect modules and supporting circuits:

- **Overdrive Module** – Op-amp gain stage with diode clipping and tone control.
- **Phaser Module** – Multi-stage all-pass filters with LFO-controlled JFETs or OTAs.
- **Tremolo Module** – LFO-driven voltage-controlled amplifier.

- **Spring Reverb Module** – Mechanical reverb tank with preamp and recovery stage.
- **Buffers and Power Supply** – Op-amp buffers and regulated dual-rail supply.
- **Switching System** – Buffered bypass or latching relays for each module.

Modules are interconnected using analog audio paths on PCB traces, with shared power rails and star-grounding to reduce noise.

Micro-product Allocation Among Group Members

Index Number	Allocated Tasks
230045X	Overdrive module, input buffer, PCB designing
230140J	Phaser module, switching, enclosure designing
230477X	Tremolo module, output buffer
230730T	Spring reverb module, power supply

References

1. Hunter, D., *Guitar Effects Pedals: The Practical Handbook*. Backbeat Books, 2004.
2. Horowitz, P., Hill, W., *The Art of Electronics*, 3rd ed. Cambridge University Press, 2015.
3. Small, R., *Op Amp Applications Handbook*. Newnes, 2005.
4. Accutronics, *Spring Reverb Tank Datasheets*, 2020.

Voice-Controlled LED Light Show

Team SuperTronics

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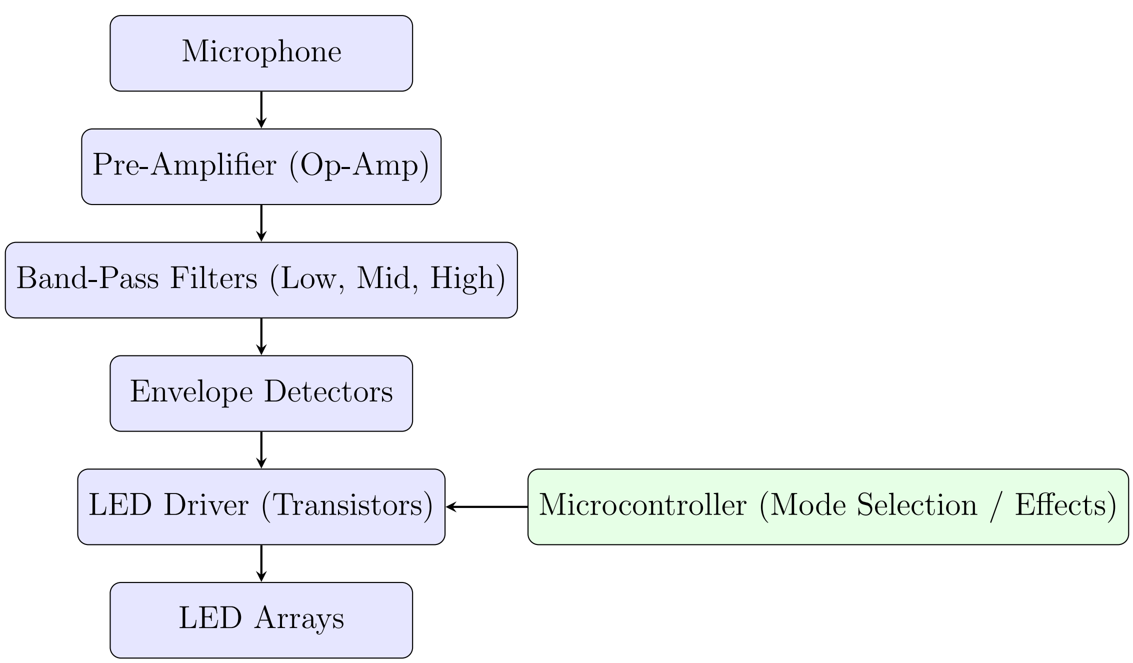
Functionality

The proposed system will create an LED light display that responds to the user's voice or music in real-time. The system will capture audio signals through a microphone, process them using analog electronics (amplifiers, filters, and envelope detectors), and drive LED arrays accordingly. The brightness and pattern of the LEDs will change with the loudness and frequency components of the input audio. Optionally a microcontroller will be used to display effects and mode selection.

Block Diagram



Methodology

The development process will be carried out in the following stages:

1. **Research and analysis.** Review existing analog audio-responsive lighting circuits, studying their topologies, performance characteristics, and integration requirements. Identify component types suitable for low-noise amplification, accurate frequency filtering, and reliable LED driving.

2. **Design stage.**

    - Draw schematics for each subsystem (Amplification, Filtering, Detection, LED Driving, Power Supply) using professional circuit design software ensuring proper impedance matching between stages to minimize signal loss
    - Incorporate transistor-based or op-amp-based driver circuits capable of handling the required LED current.

3. **Simulation.** Model each circuit in LTspice to verify gain levels, filter frequency responses, envelope detection behavior, and LED driver performance before hardware implementation.

4. **Prototyping.**

    - Build each subsystem separately on a breadboard.
    - Test the performance of each subsystem individually checking noise levels, frequency separation, detection accuracy, and LED brightness control.

5. **Integration.** Interconnect all subsystems in sequence to form the complete system. Verify that the LED arrays respond in real-time to changes in audio input across different frequency bands.

6. **PCB design.** Design the final PCB with proper grounding, signal routing, and decoupling to minimize interference between analog audio paths and high-current LED lines.

7. **Enclosure design.** Create a 3D model of the enclosure, ensuring secure mounting for the PCB, microphone, LEDs, and optional microcontroller interface, with attention to aesthetics.

8. **Assembly and testing.** Assemble all components into the pcb, connect the power supply, and tests with different audio sources to verify responsiveness, stability, and lighting effects.

Micro Products and Interconnections

1. **Audio Capturing Module:** A microphone will be used to capture the voice or music signal from the environment.
2. **Signal Conditioning Module:** The weak microphone signal will be amplified using a low-noise op-amp based preamplifier to bring it to a usable voltage level.
3. **Frequency Band Separation Module:** The amplified signal will be split into several distinct frequency ranges (low, mid, high) using analog band-pass filter circuits.
4. **Amplitude Detection Module:** Each filtered signal will pass through an envelope detector circuit to measure its amplitude, which corresponds to the loudness in that frequency range.
5. **LED Driving Module:** The detected amplitude signals will control transistor-based LED driver circuits. The brightness of each LED array will change dynamically according to the corresponding frequency band's amplitude.
6. **Optional Microcontroller Control:** A microcontroller may be used to select different LED display patterns or operating modes. The microcontroller will interface with the LED driver stage without replacing the analog processing path.
7. **Power Supply Module:** Provides regulated DC voltages required for the entire system. This includes a dedicated LED supply rail for the high-current LED arrays, and regulated low-voltage rails for the analog circuits and optional microcontroller.

End-to-end flow: Microphone → *Amplification* → *Filtering* → *Detection* → *Output* (LED arrays).

Task Allocation

Subsystem / Task	**Assigned Member**
Audio capturing and Signal conditioning modules	230045X
Frequency band seperation and optional microcontroller module	230477X
Amplitude detection and Power supply modules	230140J
LED driving and power supply modules	230730T

References

1. Texas Instruments, *Single-Supply, Electret Microphone Pre-Amplifier Reference Design (TIDU765)*: https://www.ti.com/lit/ug/tidu765/tidu765.pdf
2. Elliott Sound Products, *LED Lighting Effects for Audio*: https://sound-au.com/project60.htm

Feasibility Report
Analogue Thermoelectric Cooling Controller

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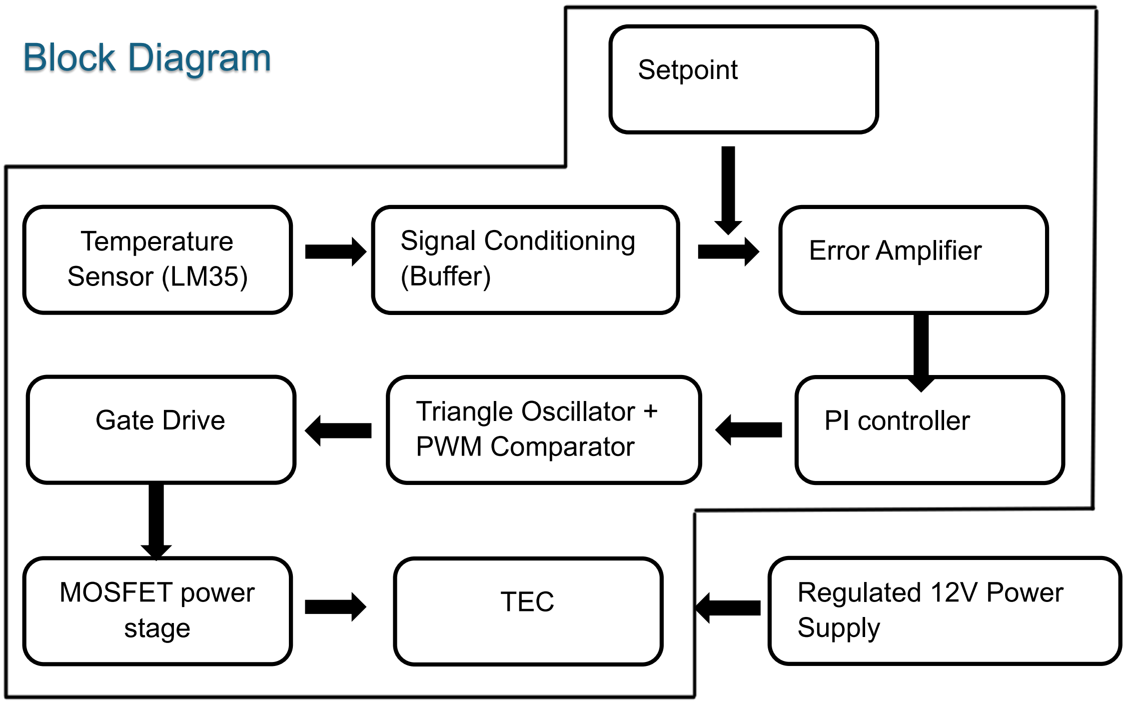
Functionality

The proposed system will control the temperature of a 12 V, 6 A thermoelectric cooler (TEC) using a fully analog feedback loop. A proportional-integral (PI) control stage, implemented using op-amps and RC networks, will regulate the TEC current via a PWM power driver stage. The system will maintain a set temperature setpoint despite environmental changes. Current sensing and overcurrent protection will ensure safe operation. A microcontroller may optionally be used for displaying the setpoint and measured temperature, but the control and actuation path will remain purely analog.

Block Diagram



Methodology

- **Research and component selection**: Study TEC operation, op-amp PI controllers, PWM generation, and MOSFET driver design.

- **Circuit design**: Draw schematics for sensing, PI control, PWM generation, MOSFET driver, and protection stages.
- **Subsystem prototyping**: Build and test each module individually on breadboards to verify correct operation.
- **Integration**: Connect all modules in sequence and verify stable temperature regulation.
- **Testing**: Measure temperature stability, response to disturbances, and overcurrent protection function.
- **Enclosure design**: Prepare housing with heatsink and fan for hot side, ensuring safe mounting of components.
- **Final assembly**: Install all components, route wiring, and finalize the project for demonstration.

Micro Products and Interconnections

- **Temperature Sensor Module**: LM35 sensor with op-amp conditioning outputting analog voltage to error amplifier.
- **Setpoint Generator**: Potentiometer with buffer stage providing reference voltage.
- **Error Amplifier + PI Controller**: Op-amp stages generating control voltage for PWM.
- **Triangle Wave Oscillator**: Op-amp integrator + comparator generating sawtooth/triangle waveform.
- **PWM Comparator**: Compares PI output to triangle wave, producing PWM signal.
- **MOSFET Power Stage**: Low-side driver switching TEC power line.
- **Current Sense Module**: Shunt resistor with op-amp amplifier to measure TEC current.
- **Overcurrent Protection Module**: Comparator output disabling PWM when threshold exceeded.
- **Display Module**: Microcontroller reads temperature and setpoint, displaying via I^2C/SPI LCD.

Micro-product Allocation Among Group Members

- Temperature sensing and setpoint generation,PCB design – 230045X
- Error amplifier and PI control design,enclosure design- 230140J
- PWM generation and triangle oscillator – 2300477X
- MOSFET power stage and current sensing – 230730T

References

1. Texas Instruments, 'LM35 Precision Centigrade Temperature Sensor Datasheet'.
2. Texas Instruments, 'LM358 Low Power Dual Op-Amp Datasheet'.
3. International Rectifier, 'IRLB3034 N-Channel MOSFET Datasheet'.
4. Horowitz, P., & Hill, W., 'The Art of Electronics', 3rd Edition.
5. Application notes on PWM current control for TEC modules.